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Assignment – II

Database Management System

(24CSH-298)

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Submitted To:

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Ques 1. Explain the following types of database security issues:

Ans: i) Legal and Ethical issues :->

Legal and ethical issues relate to laws, regulations, and moral responsibilities in handling database information.

- Privacy Protection: Personal or sensitive data (employee details, salary, medical data) must be protected.

- Data Misuse: Unauthorized use or sharing of confidential information is unethical and illegal.

- Intellectual Property Rights: Database may contain copyrighted or proprietary information.

Example: Leaking a customer personal information or using employee data without consent.

ii) Policy Issues:

Policy issues relate to the rules and guidance/guidelines established by an organization to protect it's database systems. Organisations develop security policies that defined how data should be accessed, used and protected. These policies specify user roles, access rights, responsibility of employees when handling data.

3.) System Related Issues :

System Related issues arise from technical weakness in database system infrastructure, like weak authentication systems, software - vulnerabilities improper configuration of database servers and lack of updates or security patches.

- Example: A hacker exploiting a database vulnerability to steal information.

Ques: Explain the following database security control techniques :

Ans: i.) Access Control :-

- Access control restricts who can access the database and what operations they can perform, users are given specific permission (read, write, update, delete).
- Implemented using user account roles and privilege. Ensure that only authorized users can access sensitive data.
- Eg: A clerk can view records but cannot modify them.

ii.) Inference Control :

- Inference control prevents users from deducing sensitive information from accessible data.

Eg: Preventing someone from calculating an individuals using average salary queries.

iii.) Flow control:

→ Flow control ensures that information does of flow from higher security levels to lower security levels.

→ It restricts how information moves within a system.

→ Eg: Preventing secret data from being transferred to public database.

iv.) Data Encryption:

→ Data Encryption converts plain text data into coded from cipher text to protect it from unauthorized access.

→ Eg: Encrypting credit card details stored in a database.

Ques 3: Consider a table Employee (EmpID, EmpName, Salary, DeptID). Write an SQL query to create a view that displays only EmpID, EmpName, and DeptID from the Employee table.

Ans:→ A view is a virtual table created from one or more tables. It stores a query rather than actual data.

P.T.O →

SQL Query:

```
CREATE VIEW Employee_View AS
SELECT EmpID, EmpName, DeptID
FROM Employee;
```

This View will display only EmpID, EmpName, and DeptID, hiding other columns such as Salary.